ORIUS Repo-Truthful Defense Review Dossier

Simulated 15-Reviewer / Three-Wave Audit

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Chapter 1

Purpose and Protocol

This dossier records the canonical simulated 15-reviewer audit for the live ORIUS defense package. The package is grounded in repo truth: the active three-domain lane (Battery + AV + Healthcare), the machine-generated active theorem audit, the canonical build path `make review-compile`, and the generated publication artifacts under `reports/publication/`.

ORIUS is defended as one bounded stack:

- **Phenomenon**: the Observation–Action Safety Gap (OASG),
- **Mechanism**: reliability-aware runtime safety and calibration under degraded observation,
- **Evidence lane**: Battery as witness row, AV and Healthcare as bounded defended peer rows.

The single flagship novelty sentence used across the defense package is:

> *ORIUS provides a reliability-aware runtime safety layer for physical AI under degraded observation, enforcing certificate-backed action release through uncertainty coverage, repair, and fallback.*

Novelty boundary

The novelty claim is architectural and runtime-semantic: ORIUS does not claim a new conformal method, a new robust-optimization primitive, a new universal controller, or a new conditional-coverage theorem.

Reviewer-facing separation is explicit:

- conformal methods contribute uncertainty sets, but not repaired actuation or release certification;
- runtime monitors and anomaly detectors contribute alarms or vetoes, but not tightened governed release;
- simplex and runtime-assurance schemes contribute supervisory override and fallback, but not reliability-conditioned tightening under one shared release contract; and
- safety filters and robust MPC contribute plant-local safe action geometry, but not a reusable degraded-observation release grammar across domains.

The flagship ML surface is grouped calibration plus runtime safety under degraded observation. This dossier does not defend a new conditional-coverage theorem and does not take headline novelty credit from draft theorem rows.

Every review finding must end in exactly one disposition:

- **fixed in code/tests**,
- **narrowed in prose/registers**, or
- **left open as bounded future work**.

Strict defended theorem core

The active theorem inventory is not treated as one flat ladder. The June defense package uses three public tiers:

Tier	Rows	Rule
Flagship defended	T1, T2, T3a, T4, T6, T7_Battery, T11, T_trajectory_PAC	Carries the compact defended spine and must have proof, test, witness, and bounded scope language. T7_Battery is a battery instantiation, while T11 is the universal runtime-assurance contract.
Supporting defended	T3b, T8, T11_Byzantine, T_stale_decay	Repo-truthful supporting surfaces that may be cited, but do not carry the flagship novelty or theory score alone.
Draft / non-defended	T5, T9, T10, L1–L4, T_minimax, T_sensor_converse	Visible only as scoped draft or extension surfaces and excluded from headline defended counts.

The canonical machine-readable sources for this split are `reports/publication/active_theorem_audit.*`, `reports/publication/theorem_promotion_scorecard.json`, `reports/publication/theorem_promotion_gates` `defended_theorem_core.*`, and `defended_assumption_map.*`. The raw theorem register remains an inventory, not the defense authority. For T9/T10, the generated promotion package is the sole status authority and currently keeps both rows non-promoted until all three-domain discharge gates pass.

Reviewer-role matrix

Wave	Reviewer lane	Primary attack
1	Logic / mathematics	hidden assumptions, circular dependence, false converse steps
1	Statistics / calibration	coverage semantics, grouped-coverage drift, same-data evaluation
1	Control / safety filtering	fallback semantics, repair soundness, runtime safety predicates

Wave	Reviewer lane	Primary attack
1	Systems / runtime	build integrity, artifact discipline, generated-surface drift
1	Red team / durability	vacuous success, sanitization, silent helper drift
2	ML / OQE skepticism	heuristic reliability grounding and law-vs-code mismatch
2	Information theory	Fano/Le Cam misuse and proxy bridges presented as proofs
2	Formal methods posture	traceability gate vs proof gate semantics
2	Scope / overclaim audit	defended-domain inflation and readiness inflation
2	Cross-theorem consistency	dependency drift and stale theorem inventories
3	Battery realism	witness-row assumptions and fallback/slack boundaries
3	AV realism	bounded longitudinal contract vs full-AV wording
3	Healthcare / statistical gaps	evidence tier and calibration posture
3	Reproducibility / artifacts	hash-locking, generated-asset drift, build determinism
3	Long-horizon durability	what remains defensible under repo drift

Master vulnerability register

ID	Severity	Finding	Disposition	Resolution
`V1`	critical	Build failed open on LaTeX errors	fixed in code/tests	Canonical build path now regenerates theorem audit surfaces and fails closed on theorem-surface drift.
`V2`	critical	Base `risk_bounds.py` claimed a Ville trajectory proof not in base CPU module	fixed in code/tests	Base `T_trajectory_PAC` is scoped to Bonferroni, while the Ville betting martingale E_t is implemented and Lean 4-verified in **Theorem G** (`TheoremsFGH.lean`).
`V3`	critical	Integrated theorem gate was presented as proof verification	fixed in code/tests	Gate remains traceability-release only; proof posture lives in the active audit and defended-core split.

ID	Severity	Finding	Disposition	Resolution
V4	critical	L1/L2 extension laws exceeded defensible repo truth	narrowed in prose/registers	Extension laws remain visible only as stylized draft surfaces outside the flagship theorem core.
V5	high	T_minimax / T_sensor inherited the open L1/L2 bridge	fixed in code/tests	Extension helpers now keep the converse gap open and stay out of flagship counts.
V6	high	Theorem inventory drifted from Appendix C and review language	fixed in code/tests	Active audit now emits strict theorem tiers, assumption maps, and synchronized core artifacts.
V7	high	Readiness language outran current evidence posture	narrowed in prose/registers	Defense package now treats Battery as witness row and AV plus Healthcare as bounded defended rows only.
V8	high	Base CPU domain-evidence rhetoric needed clear separation from 7-domain K-Bound	fixed in code/tests	Chapters 1–12 govern the 3-domain CPU lane, while Chapter 13 locks the 7-domain K-Bound crucible and Suites A/B hardware verification.
V9	medium	T3/T8/T9/T10/T11 remained visible as if equally defended	narrowed in prose/registers	These rows are now explicitly classified as flagship, supporting, or draft; T9/T10 status is controlled by the generated promotion package rather than by prose.
V10	medium	Assumption use was spread across prose, code, and appendices without one machine-readable map	fixed in code/tests	Defended assumption maps now expose theorem-facing assumptions and register resolution status.

Response / concession matrix

Disposition	What it means	Examples in this pass
Fixed in code/tests	Runtime semantics, theorem helpers, build path, or generated defense artifacts changed and regression checks were updated	fail-closed theorem audit build, strict defended-core split, assumption-map generation, PAC vs. Ville Theorem G separation
Narrowed in prose/registers	Manuscript/register language was reduced to match current executable and artifact truth	extension-law posture, draft theorem demotion, two-tier scope separation between Ch. 1–12 and Ch. 13
Bounded future work	Claim remains visible only as an explicit open problem, not a defended contribution	independent converse for L1/L2, repaired T10 lower bound, full-scale highway/grid statutory certification

Chapter 2

Crucible Stress Battery, K-Bound, and HIL Verification Audit

In addition to the promoted three-domain CPU runtime surfaces, the executable verification suites (`experiments/orius_foundations/crucible_and_swarm_benchmarks.py` and `experiments/orius_foundations` record the four-stage Crucible adversarial stress battery, two-regime K-Bound conformal adaptation gating, 100k GPU swarm simulations, and closed-loop robot and battery bench verification:

Verification Stage	**Stress Regime / Barrier**	**Observed Margin**	**Status**
100,000-Agent Swarm (1.0s Adv.)	Learned `AdversaryNet` (1000 ms)	+15.74 m (0/100k crashes)	Verified
100,000-Agent Swarm (3.0s Adv.)	Extreme `AdversaryNet` (3000 ms)	+55.02 m (0/100k crashes)	Verified
3D 10,000-Agent Swarm ($R_{\text{obs}} = 4$m)	True spherical surface clearance	+4.679 m (0/10k pen.)	Verified
4-Embodiment Velocity-Aware Shield	AV / 6-DoF Arm / Drone / Grid	+29.84m/ + 0.279m/ + 6.97m/0.9486pu	Verified
Crucible 1: 10,000-Cycle Fuzzing	Extreme latency, $w_t \in (0, 1]$	0 / 10,000 Faults	Passed
Crucible 2: Stealth FDI Spoofing	-12 m/s ramp spoofing bias	+38.54 m clearance	Immune
Crucible 3: Triple-Fault Cascade	520 ms lag + 40% brake + ice	+8.01 m clearance	Survived
Crucible 4: 1,000-Trial Monte Carlo	Randomized $v_0, d_0, \tau_{\text{fault}}$	+14.23 m (0/1,000 crashes)	Passed
K-Bound Two-Regime TTA Gate	0 ms shift (`ADAPT`) / 650 ms lag (`FREEZE`)	0.0312 MSE (-73.7%) / 0.4920 MSE	Verified
Native C++17 `-O3` & CAN HIL	2M steps / ISO 11898-2 500 kbps	0.0056–0.0152 μs / +4.17 m	Verified
Theorem F (`TheoremsFGH.lean`)	Unbounded blackout terminal safe-stop	$\mathcal{Z}_{\text{stop}}$ absorbing invariance (0 `sorry`)	Proved
Theorem G (`TheoremsFGH.lean`)	Ville martingale E_t + kinematic projection	$u_t \in \mathcal{A}_{\text{TV}-\text{SSK}}$ invariance (0 `sorry`)	Proved
Theorem H (`TheoremsFGH.lean`)	Financial L2 liquidity floor $L_{\text{depth}} \geq L_{\min} > 0$	Non-singular solvency (0 `sorry`)	Proved

7-Domain Real-Data Crash-Free Suite	224k steps (OPSD/Waymo/BIDMC/ADS-B/CCPP/KITTI/L2)	0/56,000 ORIUS violations (0.000% TSVR)	Verified
Suite A: Closed-Loop Mobile Robot	200 episodes (`R-1`–`R-4`, 4.0s blackout, spoof)	0/200 collisions (+0.86–4.94 m)	Verified
Suite B: Battery BMS Load Bench	4,800 steps (`B-1`–`B-3`, 50 A 2-cell bus)	0/4,800 violations (+2.07% SoC, 49.3 A)	Verified

Chapter 3

Final Disposition Summary

The canonical defense package does *not* claim that every theorem-like row in the repository is equally rigorous. It claims something narrower and more defensible:

1. the live defended program is the promoted Battery + AV + Healthcare lane,
2. the registry-canonical theorem core is explicitly classified into flagship, supporting, and draft surfaces,
3. proof rigor is tracked by the active theorem audit rather than by a traceability gate pretending to be a proof checker, and
4. any row that still lacks aligned code, tests, proofs, and bounded scope wording is demoted rather than rhetorically preserved.

That is the repo-truthful ORIUS defense posture after hardening: a smaller, auditable, reviewer-safe claim surface intended to survive hostile reading.